Project Proposal

Background. We will investigate individual differences in learning and planning in the two-step task (TST; Daw et al., 2011) using a computational model that separates learning and planning processes, and how these differences relate to anxious-depression scores. The TST is primarily used to distinguish between model-based (MB) and model-free (MF) behavior in participants (Daw et al., 2011). Differences in MB and MF behavior have been found in compulsive disorders (Gillan et al., 2016; Voon et al., 2015), but evidence in depression is sparse (Donegan et al., 2023; Patzelt et al., 2019). Logistic regression and hybrid reinforcement learning models are the most common ways of analyzing the behavior from the TST (Daw et al., 2011b). However, Feher da Silva & Hare (2020) showed via simulations that these types of analyses can classify purely MB learners as MF or hybrid if they use an incorrect mental model of the task. This implies that apparent MF behavior may reflect differences in learning or representing the task rather than differences in planning.

Dataset. We will use the data from Donegan et al., 2025: an observational study (N=781) with a control group receiving no treatment (N=87), an antidepressant group (N=83), and a therapy group receiving online cognitive behavioral therapy (N=611). All participants completed a gamified TST and self-reported mental health questionnaires summarized into three transdiagnostic scores: anxious-depression, compulsivity and intrusive thought, and social withdrawal. All data were collected at baseline and at a 4-week follow-up.

Aims. While previous studies show mixed results in model-based planning in depression, there is evidence pointing towards cognitive inflexibility and rigid prior updating in depressed individuals (Bottemanne et al., 2025; Singh et al., 2025). Therefore, we hypothesize that the observed behavior in the TST in depressed individuals might reflect differences in learning (belief-updating), rather than planning, a distinction that previous analyzes were not able to make. Hence, we aim to answer the following questions:

1. Do baseline individual differences in learning- or planning-related processes relate to self-reported depression scores?
2. Can baseline individual differences in learning- or planning-related processes predict treatment outcomes to CBT and antidepressants?
3. Is there a change from baseline to follow-up in learning or planning related parameters?

Analysis. Due to reward probabilities evolving in time as random Gaussian walks, we will use 2-level HGFs with custom, theoretically justified, planning processes. We will also include several models drawn from the reinforcement learning literature. We will perform model comparison and evaluation following the Bayesian workflow for generative modeling in psychiatry described in Hess et al., 2025. The estimated model parameters will then be related to depressive scores via multi-level Bayesian regression to answer our research questions.

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Statistical Analysis Plan

13.05.2026

Step 0: Dataset Splitting

We will make one global stratified split of the full dataset as: 80% train, 20% test. This will be stratified by group so that group proportions are preserved in both sets. We will make the split once at the very beginning before any model fitting or inspection. This will give us:

- iCBT: 489 train / 122 test
- Antidepressants: 66 train / 17 test
- Control: 70 train / 17 test

Step 01: Model Space Definition

We defined 4 models of the TST task. All models are structured into 4 main parts by temporal (1 Stage and 2 Stage) and cognitive modalities (Learning and Planning). The models are described below, with a schematic illustrations of each model. For simplicity, we use subscripts to denote to which stage each free parameter belongs to, and all free parameters are included in the schematics as dashed circles. In all models involving the HGF's, the initial mean and variance of the continuous nodes are kept fixed.

1. Model: Fixed Transition + HGFs + Canonical Bellman + Softmax

   This is our baseline model where we assume that:

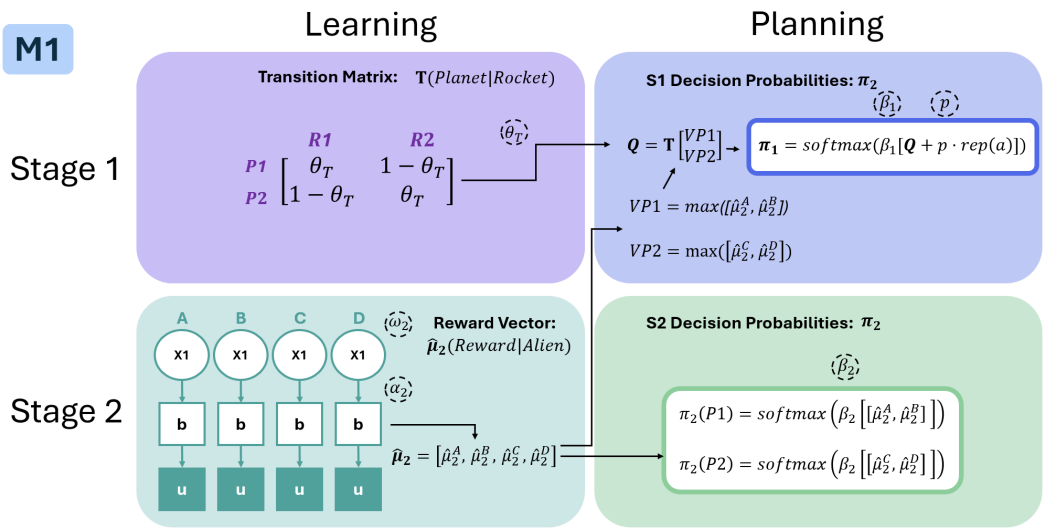
   Stage 1 - Learning: The belief about first stage transitions remains fixed, but we allow this belief to vary across participants, and therefore we will treat the transition probability θ_T to be a free parameter.

   Stage 2 - Learning: Participants form separate beliefs per each alien, and therefore we model each alien with its own 2-level HGF. Here, we assume that participants might differ in how fast they update their beliefs captured by the free parameter α_2, and that participants might differ in their baseline expectations of the volatility of reward probabilities, captured by the free parameter ω_2. We assume single shared parameter across individual HGF's.

   Stage 1 - Planning: Participants choose between the two rockets in the first stage by trying to maximize their expected future reward. We model this according to the canonical Bellman formulation, where the agent first considers the most rewarding aliens per planet and multiplies this by the learned 1-stage transition matrix to obtain the action-value $\boldsymbol{Q}$. This is then passed to the softmax function scaled by the temperature parameter β_1 to obtain a proper probability distribution over 1-stage actions. Additionally, we include a perseverence term to also capture any habitual tendencies to simply repeat last actions, where e.g. rep = [1,0] if participants chose rocket 1 at a previous timestep, scaled by the perseverence parameter p

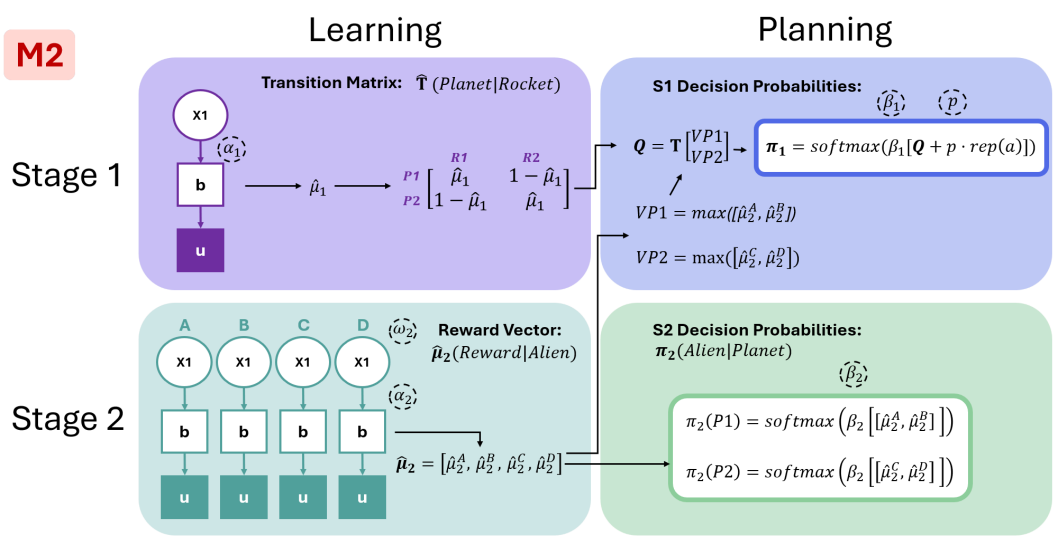
   Stage 2 - Planning: That participants probabilistically sample their 2-stage actions according to the learned reward probabilities, depending on the planet, scaled by the softmax temperature

parameter β_2.



2. Model: HGF + HGFs + Canonical Bellman + Softmax

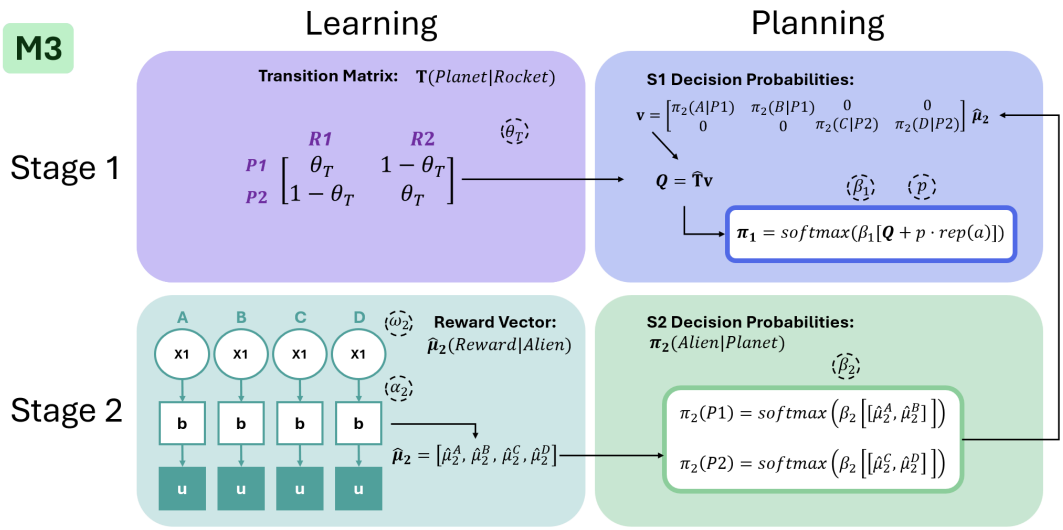
   This model is structurally similar to Model 1. , where we only replace the Stage 1 - Planning with a single 2-level HGF. The rationale behind including the HGF in the first stage is to address the concerns raised by da Silva & Hare (2020), which points out that even though participants are explicitly told that the probabilistic structure of the first stage transitions remains fixed, some participants might nevertheless keep forming beliefs about this transition throughout the task. We will only include a single free parameter in this part, namely a coupling strength α_2 to capture individual differences in the rate of belief updating about the 1-stage transition.



3. Model: Fixed Transition + HGFs + Softmax Bellman + Softmax

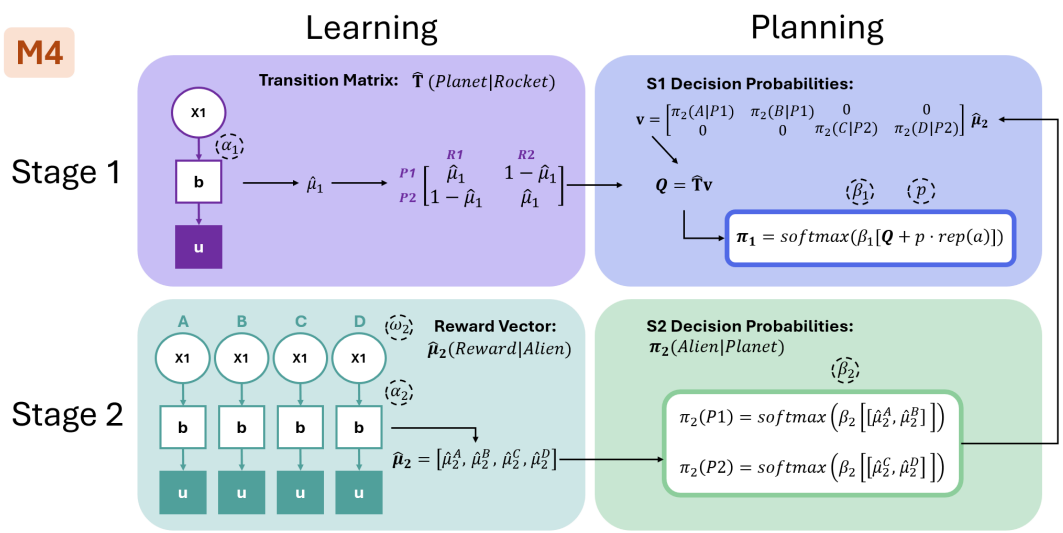
   The third model is again a variation of Model 1., but this time with a different 1-stage planning. Here, we assume that instead of looking for the best possible reward available in the state (off-policy),

participants calculate the expected value of each planet by considering the expected rewards per each alien, but scaled by its own 2-stage action probabilities (on-policy). This form of planning function is also called Bellman's Softmax Operator, or simply Bellman Softmax.



4. Model: HGF + HGFs + Softmax Bellman + Softmax

   The last model is simply an implementation of Model 3. with a single 2-level HGF learning the 1-stage transitions exactly as in Model 2.



Step 02: Prior Specifications

We will first specify initial priors on all free parameters based on theoretical considerations and existing literature. We will then perform prior predictive checks: sampling ~100 parameter vectors from the prior distributions, and simulating data with the same number of trials and transition structure as the real task. We will then inspect the evolution of simulated choices and internal belief states. Concretely, we will check that 1) simulated choices span plausible range and are not e.g. all-stay or all-

switch and 2) and the trajectories of belief states cover both extremes and intermediate values, e.g. updating very slowly or very fast, and everything in between. If any of these checks fail, we will revise priors and repeat this procedure until the simulated behavior is qualitatively consistent with what we would expect from human participants in the TST.

Step 03: Parameter Recovery

Before fitting our models to the real data, we will check that model parameters can be recovered from simulated data. Here, we again sample 100 parameter vectors from the prior for each model, and simulate full TST dataset per vector using the same trial structure as the real task. The simulated behavior will then be fitted to models to estimate the model parameters. For each model we will calculate Pearson correlation and make a visual scatter plot between generative and estimated parameter values for each parameter. We will consider a parameter recoverable if $r > 0.6$. We will note down all parameters falling below this threshold and will address this in the discussion section of our final project report.

Step 04: Model Fitting

We fill fit all models to the training set using the No-U-Turn-Sampler implemented in the Turing.jl library in Julia. We will estimate the posterior values of each parameter per each participant for each session (baseline and follow-up). We will use 4 chains, 1000 samples per chain, and will access convergence using R-Hat < 1.01. We will note down all participants where convergence fails across models.

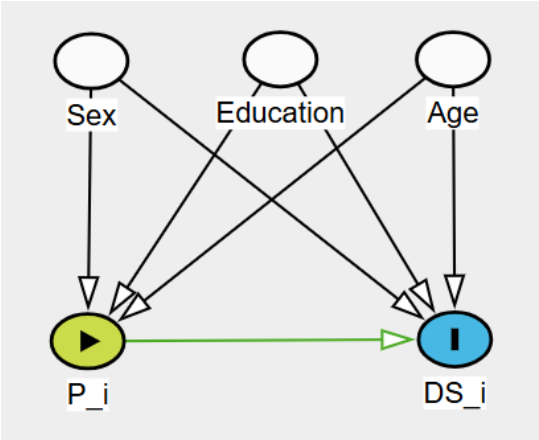
Step 05: Model Comparison

We will perform model comparison using the Pareto-smoothed importance sampling leave-one-out cross-validation (PSIS-LOO) computed on the training set. The individual-level parameter estimates from the winning model will be carried forward to the generative embedding part. Concretely, we will select the model with the highest ELPD as our winning model. In the case where two or more models fall within 2 SE of each other in terms of ELPD, we will consider them equivalent and proceed with the mechanistically simplest model, which in our case corresponds to Model 1 (Fixed Transition + HGFs + Canonical Bellman + Softmax). We will use a threshold of $k > 0.7$ of the Pareto k diagnostic for identification of problematic observations. We will report the percentage of trials exceeding this threshold. Participants for whom more than 20% of trials exceed this threshold will be noted down and included in the subsequent analyses. However, we will address this in the discussion section of the final project report to highlight that a sensitivity analysis excluding these participants would still need to be done in order to confirm that the conclusions of our analyses remain unaffected.

Step 06: Generative Embedding and Causal Modeling

To answer our research questions, we will relate individual-level parameter estimates to anxious-depression scores. Specifically, we will extract the medians from the estimated posterior distributions, which will then be used as regressors in multi-level Bayesian regression models (RQ1 and RQ3), and as a predictor in a machine learning algorithm (RQ2). We will build the regression models by considering the assumed causal relationships between variables specified via Directed Acyclic Graphs (DAGs). All analyses will be carried out in Julia and R.

RQ1 — 1. Do baseline individual differences in learning- or planning-related processes relate to self-reported depression scores (at baseline)?



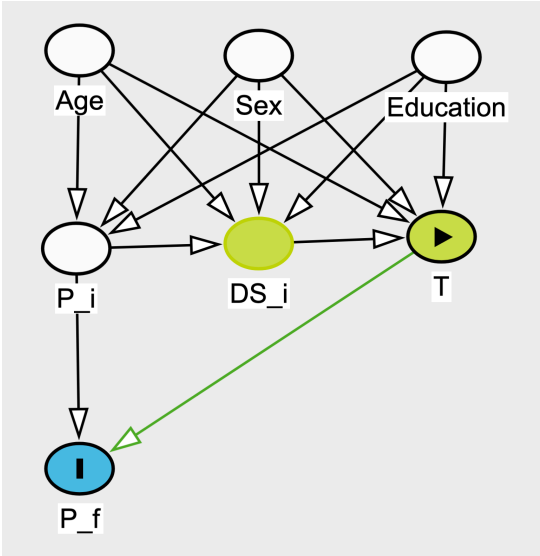
We here model the effect of baseline parameters (P_i) on baseline depression scores (DS_i). Further, we have sex, education, and age acting as confounders and we will therefore control for them by including them in our models as covariates.

RQ2 — Can Baseline Individual Differences in Learning or Planning related Processes Predict Treatment Outcomes (SSRI, iCBT)

Since this is a question considering pure predictive power of the estimated computational parameters on treatment outcomes, here we forego a Bayesian regression in favor of a machine learning approach optimized for out-of-sample predictive performance. We will use XGBoost as our primary algorithm, with the baseline individual-level parameter estimates and depression scores, treatment group, and demographic variables (sex, age, education) as predictors, and follow-up depression scores as the predicted outcome variable.

RQ3 — "What is the effect of treatment on follow-up parameter estimates (independent of initial parameter estimates)?"

We rephrased this research question from the original formulation: "Is there a change from baseline to follow-up in learning or planning related parameters?", in order to resolve a partial ambiguity regarding which variable serves as the exposure. The revised formulation explicitly identifies treatment as the exposure of interest. The assumed causal relationship used to build the regression model to estimate the effect of treatment on the follow-up parameter estimates is depicted in the following DAG, (with control variables P_i, Age, Sex, and Education):



Step 07: Generative Embedding Validation

RQ1 and RQ3 (causal)

- We will Report posterior parameter estimates and 95% credible intervals.
- The evidence for/against causal effects will be reported by the proportion of posterior mass above or below zero.

RQ2 (predictive):

- We will evaluate out-of-sample predictive performance on the held-out test set, namely:
  - R^2 and MAE overall and separately per group (iCBT, antidepressants, controls)
  - A permutation test (1000 permutations, shuffling outcome labels) to assess whether performance exceeds chance and will report the resulting p-value
- Lastly, we will report the results of the XGBoost feature importance analysis to identify which baseline parameters are most predictive of the treatment outcomes.

Possible Future Extensions:

Steps 1-5

- The cognitive modelling part of this Statistical Analysis plan implicitly tests two distinct hypotheses: whether stage 1 learning is better described static or dynamic, and wheter thage 1 planning is better described as off-policy (canonical Bellman) or on-policy (Bellman Softmax). These two questions could be addressed explicitly as research questions on their own right. Namely: 'Is the

assumed transition structure of stage 1 better described as static or dynamic?' and 'Is stage 1 planning better described as off-policy or on-policy planning?'.

- Additionally, our current model space considers two planning operators (canonical Bellman and Bellman Softmax), but a third alternative exists, known as the Bellman Mellowmax, a log-sum-exponential operator that assumes equal weighting of optimal and non-optimal actions, and considers the total expected reward of each “Planet”. We did not include this model due to time-constaints, however, future work could involve extending the model space to include Mellowmax, which would make the model comparison analysis more comprehensive.

Step 1

- The models in the current analysis which include the HGF’s treat the initial mean and variance of the continuous nodes as fixed parameters. However, these two parameters could in principle be also estimated from the observed behavior. Future work could consider estimating those parameters as well, which would allow to answer questions such as whether participants start with an initial “bias” (mean) about the true reward probabilities, or with an initial degree of “uncertainty” (variance) about the reward probabilities. These parameters are not estimated in the current analysis due to the risk of model overparameterization, and was therefore left as a direction for future work.

Step 5

- The current analysis plan also includes participants with suboptimal k-diagnostic in the Step 06 of the analysis. This decision was made due to time constraints of the current project and future analysis should exclude those subjects from the Step 06 and run sensitivity analyses to see whether this correction leads to different conclusions.

Exploratory Analyses

- Since the main objective of the current project is to analyze the relationship between individual-level computational parameters and anxious-depression, the remaining two transdiagnostic scores: compulsivity & intrusive thought, and social withdrawal, are left for future exploratory analyses.